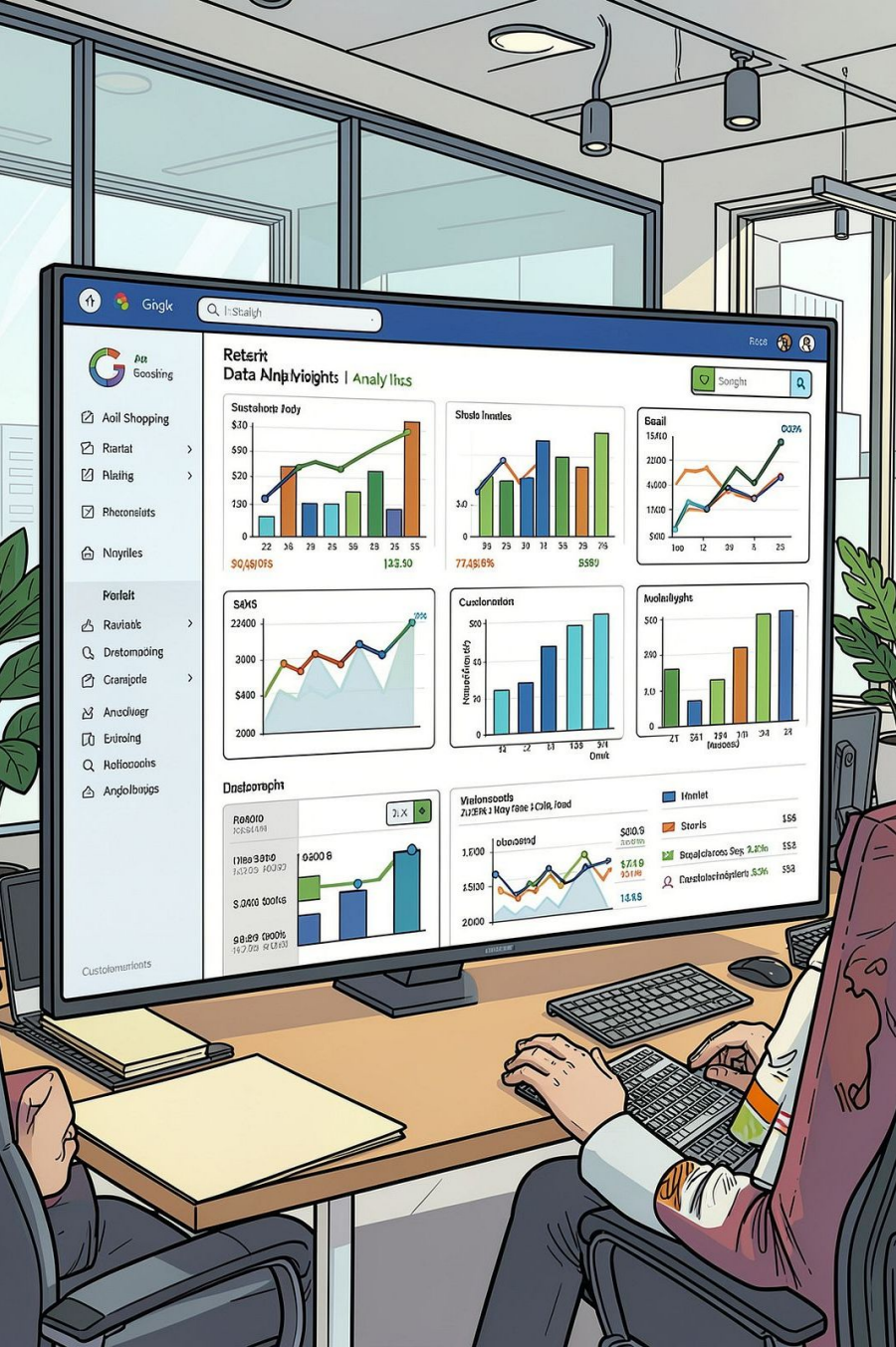




Customer Shopping Behavior Analysis

Uncovering spending patterns, customer segments, and product preferences from **3,900 transactions** to guide strategic decisions.

Dataset at a Glance

3,900

Transactions

18

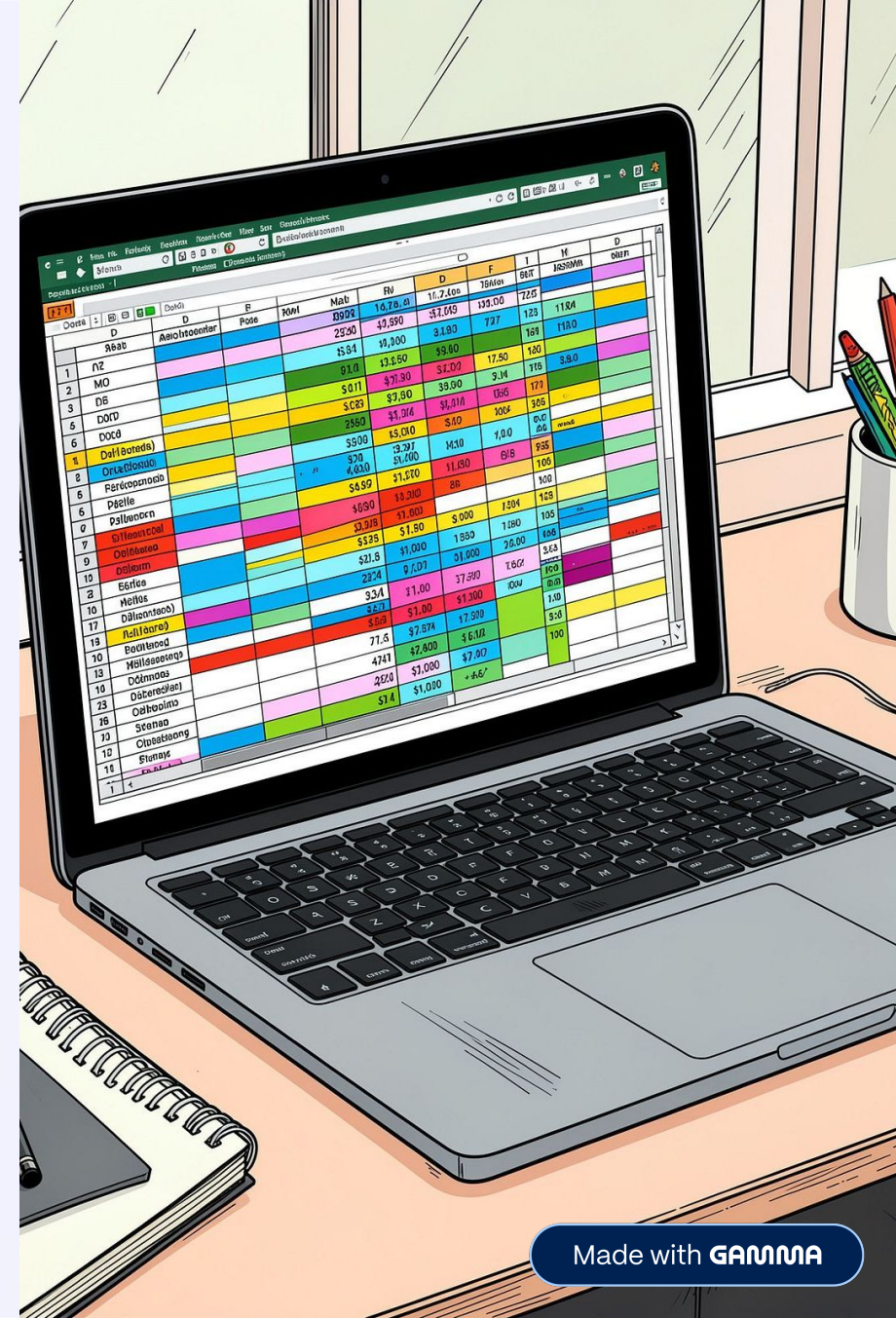
Features

37

Missing Values

Review Rating column

Key features include demographics, purchase details, and shopping behavior.



Data Preparation in Python



→ Clean & Load

Imported via pandas; checked structure with `df.info()`

→ Impute Missing Data

Median rating per category for Review Rating

→ Feature Engineering

Created `age_group` and `purchase_frequency_days`

→ PostgreSQL Integration

Loaded cleaned data for SQL analysis

Revenue by Gender

Male

\$157,890 total revenue

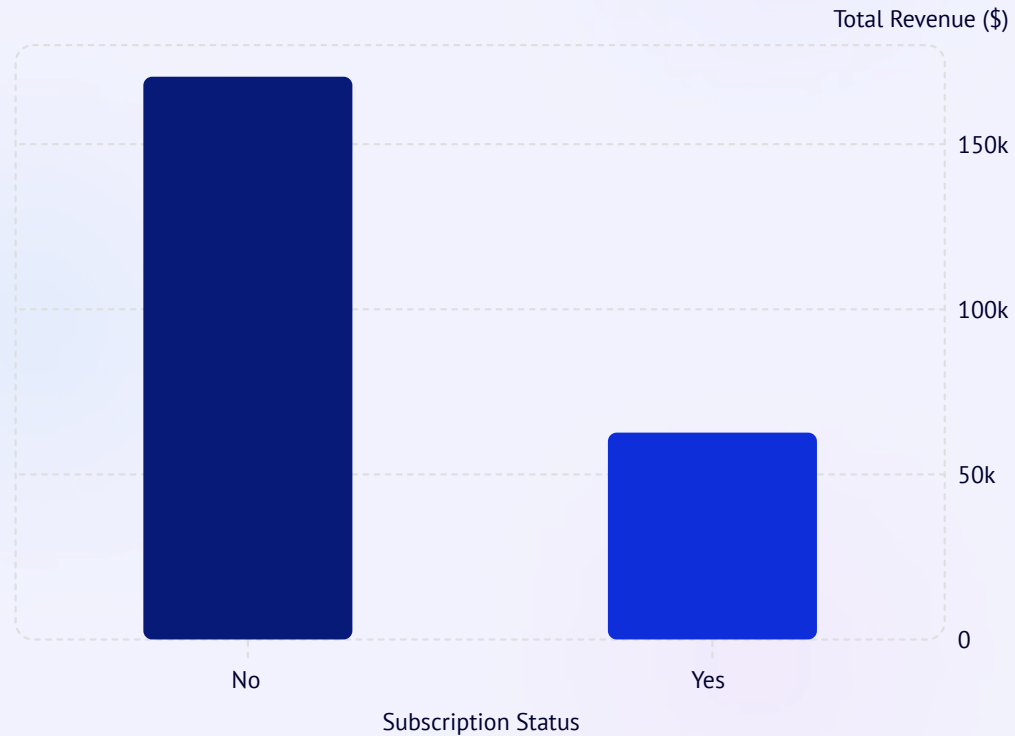
Female

\$75,191 total revenue

Male customers generated **2.1x more revenue** than female customers.



Subscribers vs. Non-Subscribers



Non-Subscribers

2,847 customers · Avg
spend: \$59.87

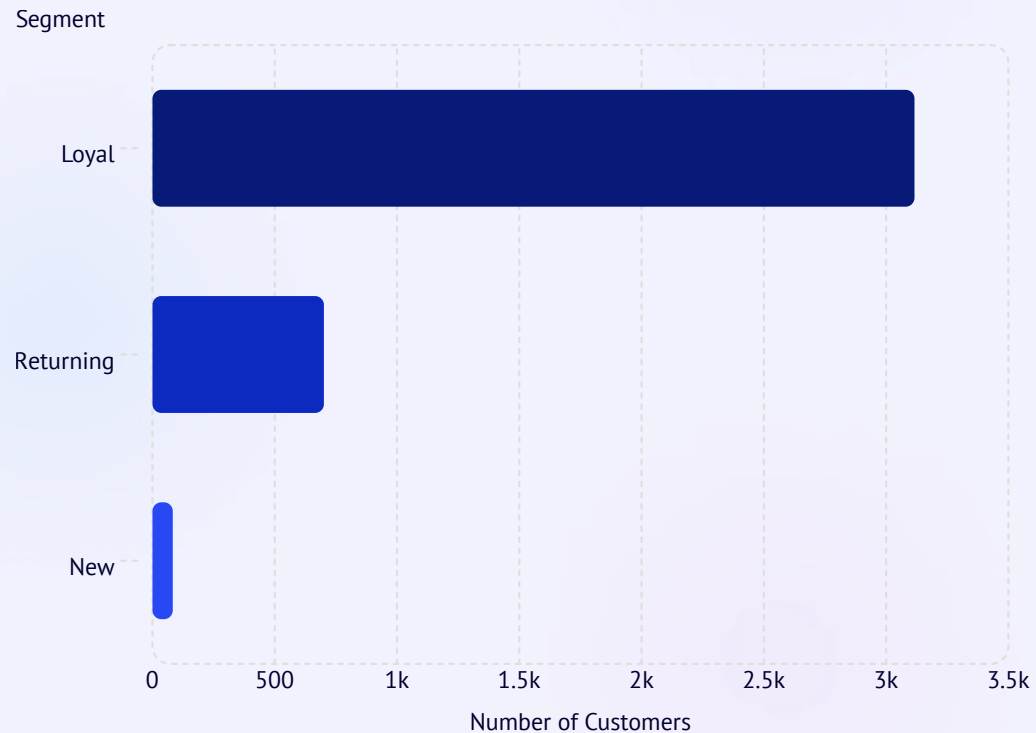
Subscribers

1,053 customers · Avg
spend: \$59.49



Non-subscribers drive **73%** of total revenue despite similar average spend.

Customer Segmentation



Segment Breakdown

Loyal — 80%

3,116 customers

Returning — 18%

701 customers

New — 2%

83 customers

Top Products by Category



Accessories

Jewelry · Sunglasses · Belt



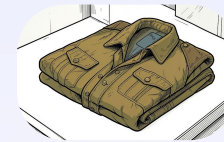
Clothing

Blouse · Pants · Shirt



Footwear

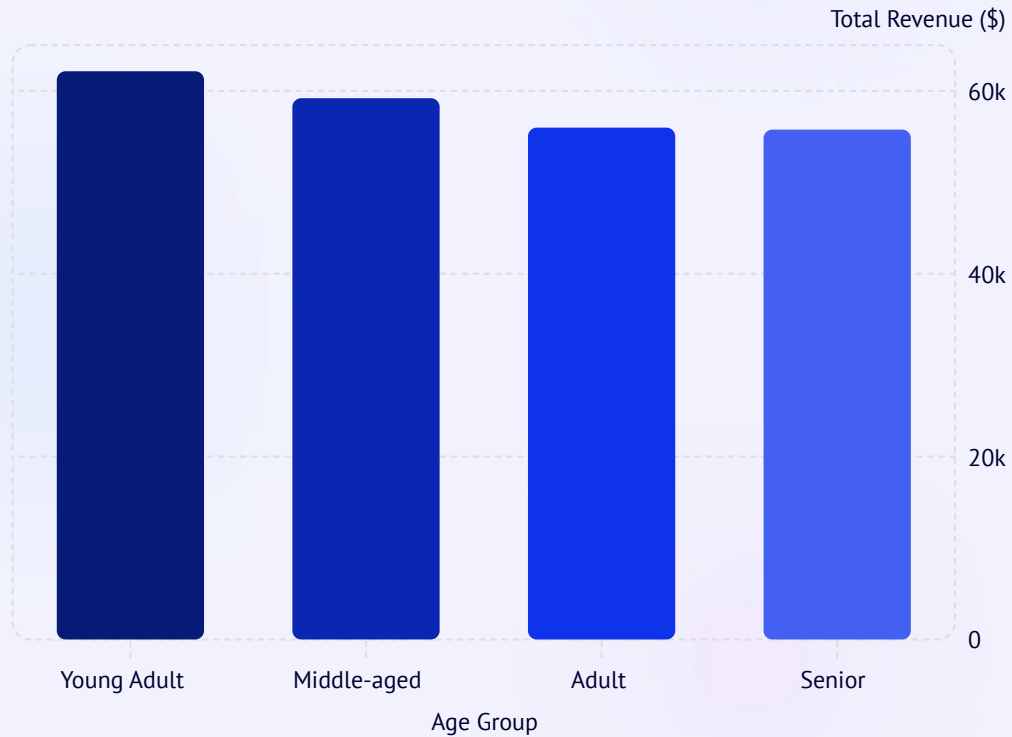
Sandals · Shoes · Sneakers



Outerwear

Jacket · Coat

Revenue by Age Group



Key Insight

Young Adults lead revenue at **\$62,143**, but all age groups contribute fairly evenly.

Avg Purchase

\$59.76

Avg Rating

3.75 / 5

Power BI Dashboard

3.9K Customers

\$59.76 Avg Purchase

3.75 Avg Rating

Interactive Filters

Gender, Category, Shipping, Subscription



Business Recommendations



Boost Subscriptions

Promote exclusive subscriber benefits



Loyalty Programs

Reward repeat buyers to grow Loyal segment



Review Discount Policy

Balance sales lift with margin control



Targeted Marketing

Focus on high-revenue age groups & express-shipping users